Korea Technology and Information Promotion Agency for SMEs

*(max 10 pages)*

# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

| **Proposer** | **Name** | Deok-Ho Kim | **Email** | tipa@jhu.edu |
|---|---|---|---|---|
| **Institute Name** | Johns Hopkins University School of Medicine | | | |
| **Address** | Rm. 724, Ross Research Building 720 Rutland Avenue Baltimore, MD 21205 | | | |

## 2. Technology Overview

| | | | | |
|---|---|---|---|---|
| **Title of Proposed Technology** | Defined, Non-Animal-Derived Three-Dimensional Scaffold for Brain Organoid-Based Drug Discovery | | | |
| **Korea's 12 National Strategic Technology Fields** | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| **Industrial Technology Classification** [1)] | **Technology Field 1 (**Large Category**)** | | **Technology Field 2 (**Large Category**)** | |
| | Bio Medical | | Mechanical materials | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Medical Bio | Pharmaceutical Bio-based Technologies and Systems | Nano/Micro Mechanical Systems | Miniature Processing/ Assembly/ Measurement Technologies |
| **Concept of Proposed Technology** | Brain organoids have emerged as advanced in vitro models capable of reproducing three-dimensional brain tissue architecture and cell–cell interactions. Their applications in brain cancer research, neurological disease modeling, and drug evaluation continue to expand. However, existing culture systems predominantly rely on animal-derived extracellular matrices (ECMs), which present inherent limitations in reproducibility and standardization due to undefined composition and batch-to-batch variability.<br><br>This proposal introduces a culture framework based on **defined, non-animal-derived three-dimensional scaffolds/matrices** composed of bio-based or advanced functional materials. These systems enable quantitative control of mechanical properties and improved reproducibility, thereby providing a more stable and controllable microenvironment for brain organoid formation and maintenance.<br><br>The joint research will explore the applicability of defined 3D scaffolds to both normal brain organoids and brain cancer organoid models. As a representative implementation, the project includes collaborative research utilizing plant-derived cellulose nanofibers (CNF) as a defined ECM scaffold. The feasibility of CNF-based matrices as alternatives to animal-derived ECMs will be | | | |

| | |
|---|---|
| | evaluated, with emphasis on standardization, reproducibility, and applicability for drug efficacy and toxicity assessment in neurological and brain cancer research. |
| **Rationale for Joint R&D Project** | Brain organoid-based drug evaluation requires the integration of advanced materials engineering with disease-specific biological modeling. Establishing reproducible and quantitative three-dimensional culture environments and applying them effectively to brain cancer research exceeds the technical capacity of a single institution or country.<br><br>This proposal therefore seeks international joint R&D collaboration between domestic teams specializing in defined 3D matrix design and overseas research groups with expertise in neuroscience and brain cancer research. Such collaboration will enable higher levels of technical validation and global applicability than could be achieved independently, and will support the development of a brain organoid-based drug evaluation framework suitable for international research environments. |
| **Key Words** | 1. Defined, non-animal-derived ECM scaffold<br>2. Brain organoid culture platform<br>3. Matrigel-alternative 3D matrix<br>4. Advanced bio-material-based, reproducible 3D microenvironment<br>5. Brain cancer drug discovery and evaluation |

*(max 10 pages)*

## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level ( 3 ) | End of the Project | Level ( 6 ) |
|---|---|---|---|---|
| **R&D Objectives** | The objectives of this project are:<br><br>1. To evaluate the applicability of defined three-dimensional scaffolds/matrices for supporting brain organoid formation and maintenance with improved reproducibility and controllability compared to animal-derived ECM systems.<br>2. To apply defined 3D culture environments to normal and brain cancer organoid models and examine their feasibility as quantitative in vitro drug evaluation platforms.<br>3. To conduct representative collaborative research using plant-derived cellulose nanofibers (CNF) as a defined ECM scaffold and assess their potential for standardized organoid culture and drug discovery support. | | | |
| **R&D Target Specifications** | **(1) Scaffold/Matrix Properties**<br><br>• Definition of mechanical property ranges (e.g., rheological characteristics) relevant to brain tissue<br>• Evaluation of diffusion and mass transport supporting cell viability<br>• Establishment of QC parameters (purity, viscosity, endotoxin levels) and acceptable variability ranges<br>  ○ mean nanofiber diameter (AFM-based): ≤ 10 nm<br>  ○ lot-to-lot variation of mean diameter: ≤ ±20 % CV<br>  ○ residual ions purity (e.g. $ClO_3^-$, $Br^-$, $NO_3^-$ as applicable): ≤ 1.0 ppm<br>  ○ endotoxin (LAL assay): ≤ 1.0 EU/mL<br>  ○ permeability (e.g. 20-70 kDa dextran): ≥ 200 – 500 % relative to animal-derived ECM<br><br>**(2) Normal Brain Organoid Formation**<br><br>• Quantitative indicators of organoid formation efficiency and morphology<br>• Biological indicators for viability and neural differentiation<br>• Reproducibility and variability control<br>• Comparative analysis versus animal-derived ECM systems<br>  ○ organoid formation efficiency: ≥ 70 – 80 % of stable 3D organoids<br>  ○ cell viability: ≥ 70 % viable cells at defined culture timepoints<br>  ○ neural identity: detection of ≥ 3 biomarkers by immunostaining and/or qPCR<br>  ○ intra-batch variability: ≤ ±30 % CV of key readouts (e.g., organoid size, viability, marker expression)<br><br>**(3) Brain Cancer Organoid Models**<br><br>• Quantitative indicators for proliferation and invasion behavior<br>• Consistency across repeated experiments<br>• Comparative analysis with animal-derived ECM models<br>  ○ proliferation markers: detection of ≥ 2 cancer proliferation biomarkers by immunostaining and/or qPCR<br>  ○ intra-experiment consistency: comparable proliferation and invasion trends across ≥ 3 independent experiments | | | |

| | |
|---|---|
| | ○ intra-experiment variability: ≤ ±30 CV of key readouts (e.g., proliferation and invasion metrics)<br><br>**(4) Drug Responsiveness Platform**<br><br>• Measurement of dose–response, IC50, and AUC values<br>• Comparison with 2D and animal-derived ECM systems<br>• Definition of dataset components and metadata structures<br>○ $IC_{50}$/AUC: for ≥ 80 % of tested compounds, $IC_{50}$ and AUC values will be reported with acceptable dose–response curve fitting (e.g., $R^2 \geq 0.8$)<br>○ intra-experiment consistency: comparable drug response trends across ≥ 3 independent experiments<br>○ intra-experiment variability: ≤ ±30 CV of key drug response parameters (e.g., $IC_{50}$, AUC)<br><br>**(5) Standardization and Scalability**<br><br>• Definition of standard specifications, SOPs, and reference standards<br>• Common criteria for culture conditions and data acquisition<br>• Applicability to multi-institutional collaborative research<br>○ SOPs: establishment of ≥ 3 standardized SOPs covering scaffold preparation, organoid culture, and drug response assays<br>○ inter-laboratory transferability: successful transfer and implementation of standardized protocols between collaborating institutions, with comparable performance outcomes |
| **R&D Plan** | **Phase 1:** Design and optimization of CNF-based hydrogel matrices with tunable physical properties<br><br>• Establish formulation design space<br>• Measure rheological properties<br>• Screen candidates for transport and stability<br><br>**Deliverables:**<br><br>• CNF hydrogel data sheets<br>• Defined material property window<br>• Draft QC criteria<br><br>**Phase 2:** Establishment of reproducible normal brain organoid models<br><br>• Optimize culture conditions<br>• Establish quantitative evaluation metrics<br>• Assess batch-to-batch variability<br><br>**Deliverables:**<br><br>• Standardized protocol<br>• Quantitative analysis framework<br>• Variability assessment report<br><br>**Phase 3:** Development of brain tumor organoid models and benchmarking<br><br>• Develop tumor organoids |

| | |
|---|---|
| | • Quantify growth and invasion<br>• Perform neutral benchmarking<br><br>**Deliverables:**<br><br>• Tumor organoid protocol<br>• Quantitative datasets<br>• Comparative evaluation summary<br><br>**Phase 4:** Validation of organoid-based drug evaluation<br><br>• Apply standard drug panels<br>• Generate dose–response metrics<br>• Organize structured datasets<br><br>**Deliverables:**<br><br>• Drug response datasets<br>• Platform validation report<br><br>**Phase 5:** Standardization and definition of reference materials and SOPs<br><br>• Define control parameters<br>• Establish QC and acceptance criteria<br>• Compile SOP package<br><br>**Deliverables:**<br><br>• ECM property criteria<br>• Final QC panel<br>• SOP and reference standard documents |

## 4. R&R (Roles and Responsibilities)

| | |
|---|---|
| **Expected R&R of Research Institute** | • Lead conceptual design and experimental framework<br>• Conduct organoid formation and drug response analysis<br>• Interpret results and manage technical integration<br>• Coordinate with international partners |
| **Expected R&R of Korean SME Partner** | • Evaluate reproducibility and implementability<br>• Assess manufacturing and process feasibility<br>• Establish batch consistency and QC conditions<br>• Provide pilot-scale validation and technical feedback |

*(max 10 pages)*

## 5. Expected Outcomes/Impact

| | |
|---|---|
| **Expected Outcomes and Impact** | 5.1 Research Outcomes<br>• Demonstration of defined ECM scaffold concepts<br>• Establishment of reproducible brain organoid models<br>• Proposal of quantitative drug evaluation frameworks<br><br>5.2 Technical Impact<br>• Address reproducibility limitations in organoid research<br>• Reduce experimental variability in drug evaluation<br>• Enable standardized international research frameworks<br><br>5.3 Commercialization and Utilization<br>• Applicability as early-stage drug evaluation platform<br>• Potential use by pharmaceutical and research institutions<br>• Extension to research-use kits and services<br><br>5.4 Industrial and Economic Impact<br>• Expansion of international collaborative research<br>• Enhancement of competitiveness in organoid technologies<br>• Foundation for technology self-reliance<br><br>5.5 International Collaboration Impact<br>• Increased technical credibility<br>• Opportunities for multinational follow-up programs<br>• Improved global recognition of domestic research capabilities |

## 6. References

(1) Adlakha YK. Human 3D brain organoids: steering the demolecularization of brain and neurological diseases. Cell Death Discov. 2023;9:221.

(2) Aisenbrey EA, Murphy WL. Synthetic alternatives to Matrigel. Nat Rev Mater. 2020.

(3) Heo JH, Kang D, Seo SJ, Jin Y. Engineering the extracellular matrix for organoid culture. Int J Stem Cells. 2022;15(1):60–69. doi:10.15283/ijsc21190

(4) Curvello R, Kerr G, Micati DJ, Chan WH, Raghuwanshi VS, Rosenbluh J, Abud HE, Garnier G. Engineered plant-based nanocellulose hydrogel for small intestinal organoid growth. Adv Sci (Weinh). 2020;8(1):2002135. doi:10.1002/advs.202002135

(5) Boerstler T, Kachkin D, Gerasimova E, Zagha N, Furlanetto F, Nayebzade N, et al. Deciphering brain organoid heterogeneity by identifying key quality determinants. Commun Biol. 2025;8:1412.

(6) Lenin S, Ponthier E, Scheer KG, Yeo ECF, Tea MN, Ebert LM, et al. A drug screening pipeline using 2D and 3D patient-derived in vitro models for pre-clinical analysis of therapy response in glioblastoma. Int J Mol Sci. 2021;22(9):4322. doi:10.3390/ijms22094322

(7) Lou L, Wang K, Cai C. Minimum information about organoid research (MIOR): a framework for reproducibility, interoperability, and translation. Nat Rev Methods Primers. 2024.

*(max 10 pages)*

(8) Tamo AK. Nanocellulose-based hydrogels as versatile materials with interesting functional properties for tissue engineering applications. J Mater Chem B. 2024;12:7692. doi:10.1039/D4TB00397G

(9) Lancaster MA, Renner M, Martin CA, Wenzel D, Bicknell LS, Hurles ME, et al. Cerebral organoids model human brain development and microcephaly. Nature. 2013;501(7467):373–379. doi:10.1038/nature12517

(10) Jacob F, Salinas RD, Zhang DY, et al. A patient-derived glioblastoma organoid model and biobank recapitulates inter- and intra-tumoral heterogeneity. Cell. 2020;180(1):188–204.e22. doi:10.1016/j.cell.2019.11.036

(11) Calandrini C, Drost J. Normal and tumor-derived organoids as a drug screening platform for tumor-specific drug vulnerabilities. STAR Protoc. 2022;3(1):101079. doi:10.1016/j.xpro.2021.101079

(12) Zhao Z, Chen X, Dowbaj AM, et al. Organoids. Nat Rev Methods Primers. 2022;2:94. doi:10.1038/s43586-022-00174-y